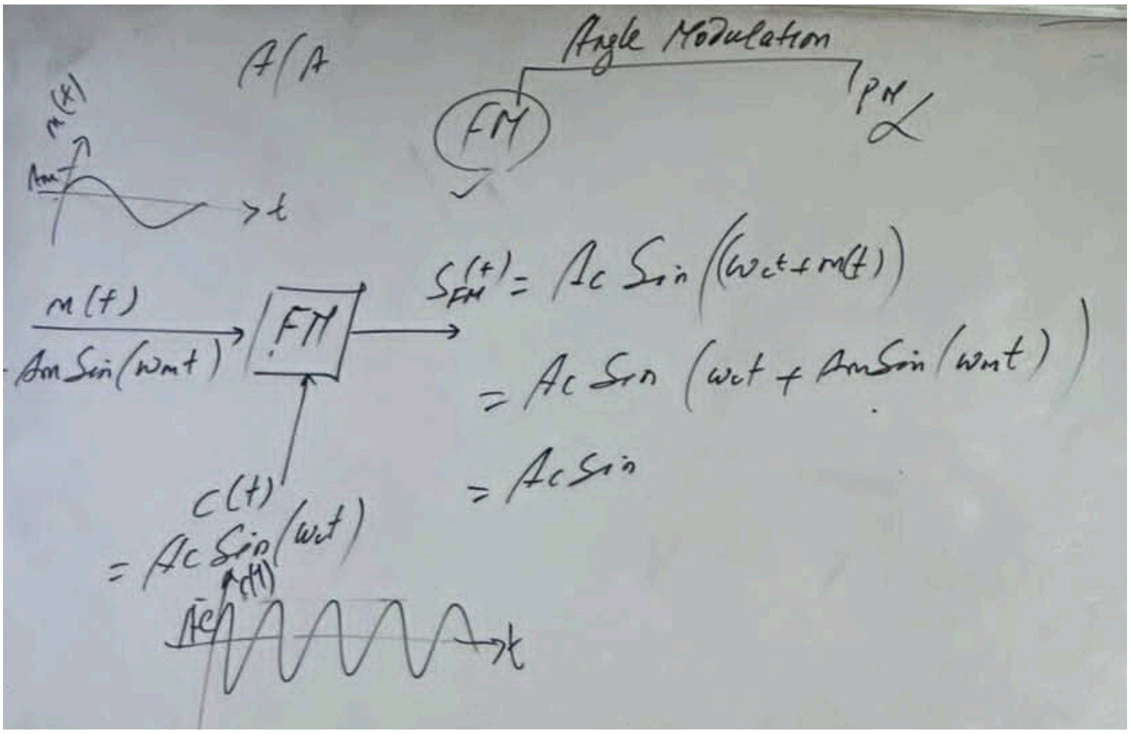
Day 8 Of Communication System Class



Frequency Modulation :

In FM, the frequency of a carrier signal is varied by the modulating signal, while the amplitude and phase of the carrier remains constant.

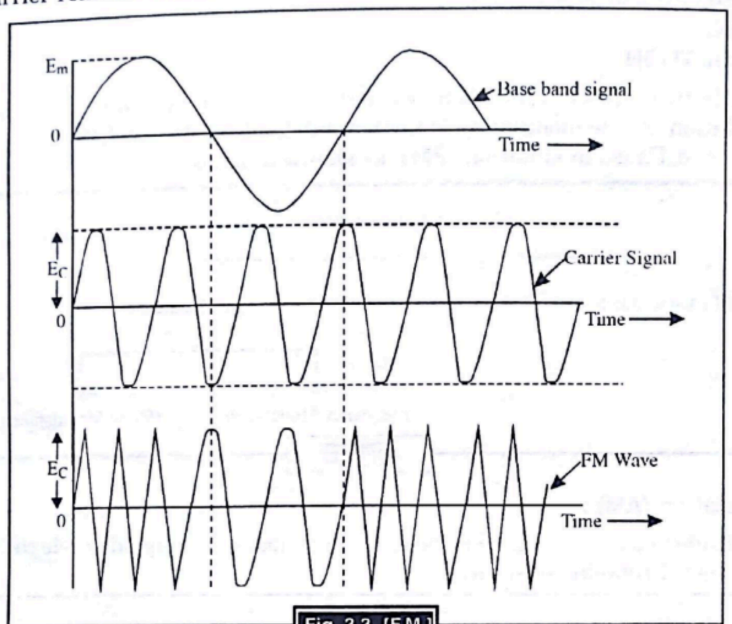


Fig. 3.3 [F.M.]

Phase Modulation :

In PM, the phase of a carrier signal is varied by the modulating signal, while other parameter remains constant.

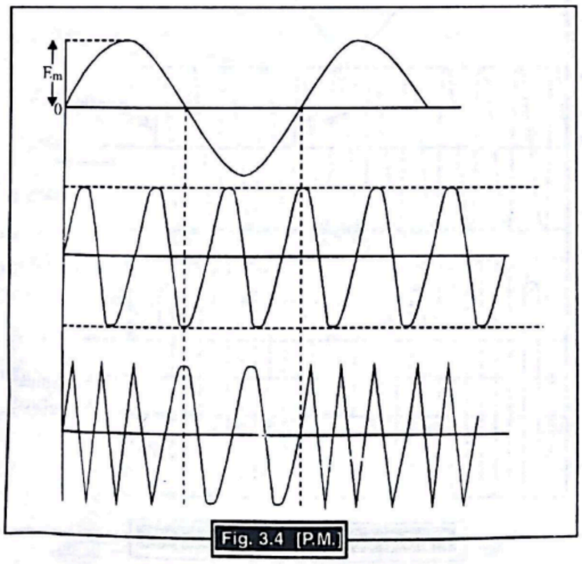


Fig. 3.4 (P.M.)

[IMP] Phase Modulation (PM) Is Not Important As Per EXAM

Frequency Modulation (FM)

FREQUENCY MODULATION (FM) :

In frequency modulation, the instantaneous frequency of the radio frequency carrier wave is varied in accordance with the signal (information) to be modulated on the wave, while the amplitude of RF wave is held constant. A frequency modulated carrier contains no variations in amplitude. All information is represented by the changes in the carrier frequency.

1. Frequency Modulation (FM)

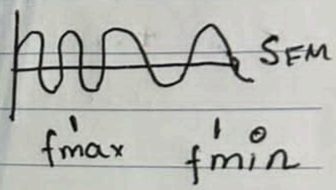
m(t)

m(t) → FM → $S_{FM} = A_c \sin\{(\omega_c + m(t))\,t\}$

$A_m \sin(\omega_m t)$

$c(t) = A_c \sin(\omega_c t)$

c(t)



* FM is clean Cause Amplitude is fixed. So, No Noise Effect is Observed.

Advantages of FM – Since noise or static interference can only amplitude modulate the carrier, noise modulation will not affect the intelligence on the carrier. Also at the receiving end, since FM detector responds only to frequency variations, and not to amplitude variations; noise will not affect the fidelity of the FM signal.

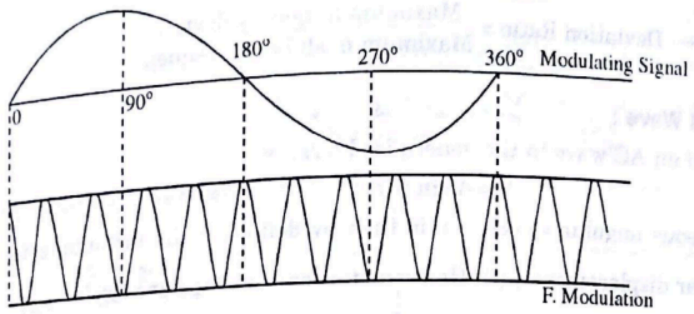


Fig. 3.13. F.M. Signal

As the value of modulating signal is reduced to zero at 180°, the carrier frequency decreases and returns to its centre value. When the modulating signal varies from zero to its negative peak at 270°, the carrier frequency to its lowest value. Finally when the modulating signal becomes again zero to 360°, the carrier is back to its centre frequency.

As the modulating signal amplitude varies, the carrier frequency varies above and below its normal centre frequency.

The amount of change in carrier frequency produced by the modulating signal is known as the Frequency Deviation or Frequency Shift.

The frequency of the modulating signal determines the rate at which the frequency deviation takes place.

For example, if a 1 KHz signal is used to modulate a carrier wave, the carrier will swing to its

Modulation index (m_f) :

$$m_f = \frac{\text{Frequency deviation}}{\text{Modulating frequency}}$$

$$\boxed{m_f = \frac{\Delta f}{f_m}}$$

$$\text{Deviation Ratio} = \frac{\text{Maximum frequency deviation}}{\text{Maximum modulating frequency}}$$

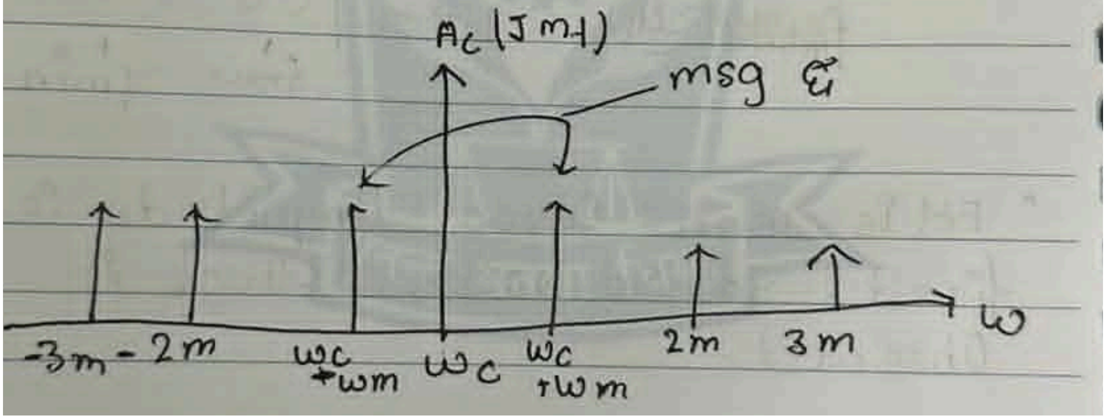
Q. Write About FM With Necessary Derivation And Figures .
[EXAM]

** wave भित्री wave → n^{th} harmonics

→ m_f = modulation index $= \dfrac{\Delta f}{f_{max}}$

↳ message frequency

$$e = A [J_0\, m_f \sin \omega_c t + J_1\, m_f (\sin (\omega_c t + \omega_m))$$
$$- \sin (\omega_c - \omega_m) t]$$
$$+ J_2\, m_f (\sin (\omega_c + 2\omega_m) t + \sin (\omega_c - 2\omega_m))$$
$$+ \ldots\ldots + n^{th} \text{ harmonics}$$



$$e = A \sin [\omega_c t + m_f \sin \omega_m t] \quad \ldots (4)$$

where m_f is the modulation index.

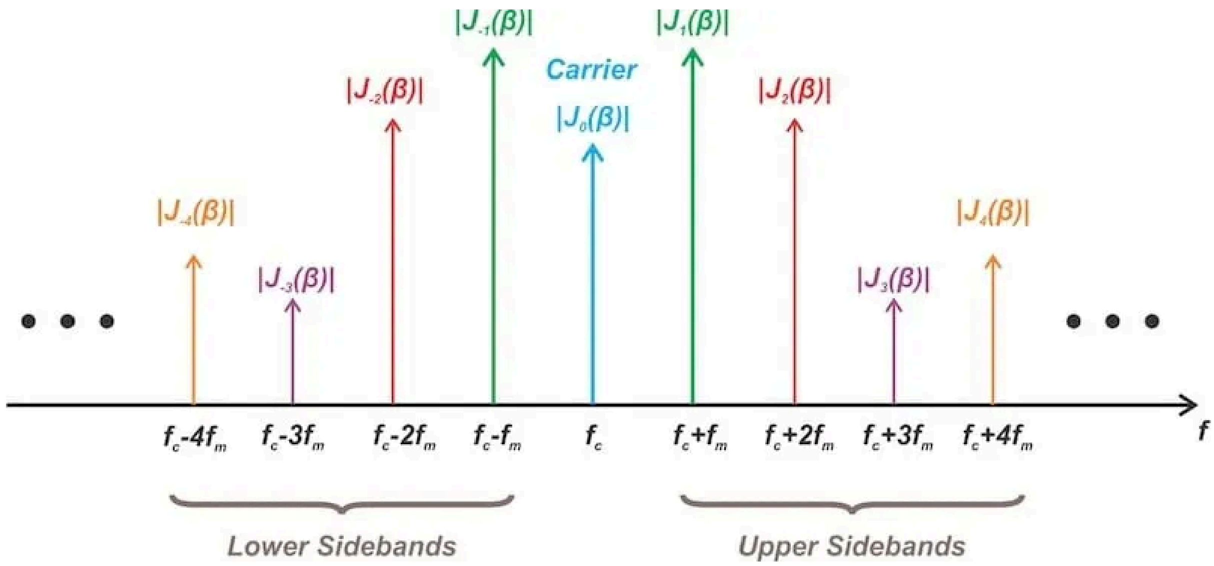
Eq. (4) is the sine of sine, the only solution involves the use of **Bessel function.**

Using Bessel function, it may be shown that the equation for FM

$$e = A \sin (\omega_c t + m_f \sin \omega_m t)$$

can be expanded to obtain

$$e = A \{J_0 (m_f) \sin \omega_c t + J_1 (m_f) [\sin (\omega_c + \omega_m) t - \sin (\omega_c - \omega_m) t]$$
$$+ J_2 (m_f) [\sin (\omega_c + 2\omega_m) t + \sin (\omega_c - 2\omega_m) t] + J_3 (m_f) [\sin (\omega_c + 3\omega_m) t + \sin (\omega_c - 3\omega_m) t]$$
$$+ J_4 (m_f) [\sin (\omega_c + 4\omega_m) t - \sin (\omega_c - 4\omega_m) t] + \ldots\ldots\ldots\ldots \}$$



Q. Write Properties Of Bessel Function . **[PYQ]**

Ans.

The following observations are made with Bessel function

1. In AM, only three frequencies are present. However, FM has an infinite number of sidebands appears. The sidebands are separated from the carrier by f_m, $2f_m$, $3f_m$ ...... etc.
2. The sideband distribution is symmetrical about the carrier frequency i.e. sidebands at equal distances from the carrier frequency have equal amplitudes.
3. Some of the Bessel coefficients may have negative values, indicating a 180° phase change for that particular pair of sidebands.
4. How many sideband components have significant amplitudes is dependent on the modulation index.
5. As modulation index increases, the value of the particular J coefficient increases. Then, assuming that the frequency deviation is constant, the relative amplitude of distant side-bands will increase as the modulation frequency is lowered.

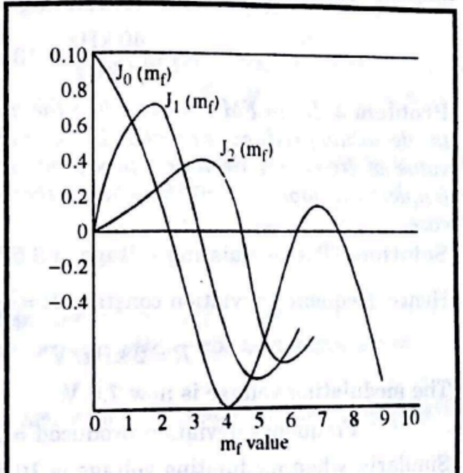
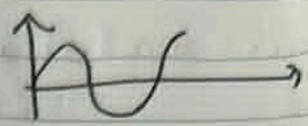


Fig. 3.15

6. In AM, the sideband power and the total transmitted power increases with the depth of modulation. However, in FM the total transmitted power always remains constant. The increased depth of modulation requires more bandwidth for transmission.

Digital to Analog (DA) — 7.1

Bit → DA → 

DA

ASK | FSK | PSK

ASK → QAM ← PSK

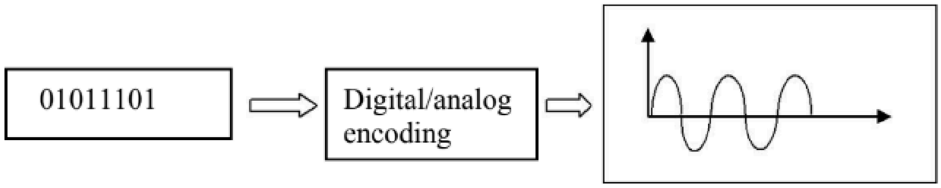
Quadrature Amplitude Modulation

A, F, P = Amplitude, frequency, phase
SK = Shift keying

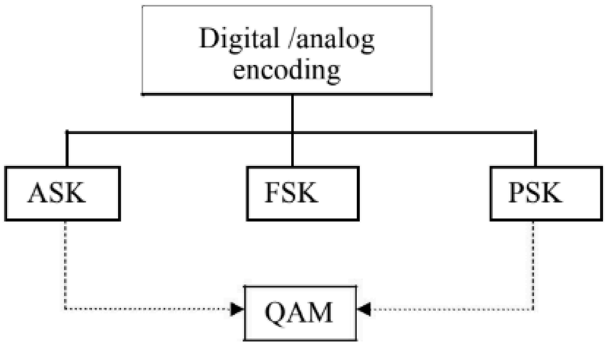
1. ASK
2. FSK
3. PSK → Binary PSK, → m-ary psk — 7.2
4. QAM (extra)

* Consollation Diagram

Digital-to-Analog Encoding



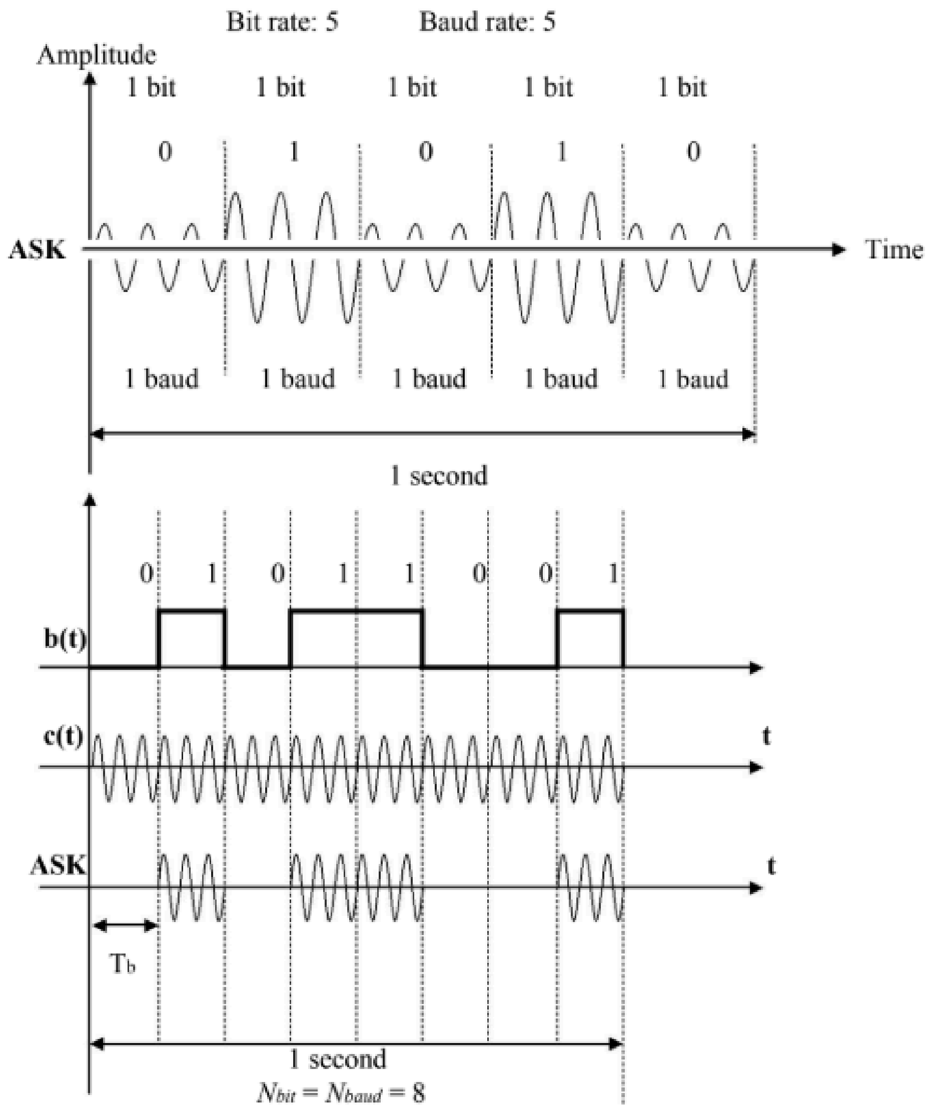
Digital-to-analog encoding is the representation if digital information by an analog signal.



Quadrature Amplitude Modulation

Amplitude Shift Keying (ASK)

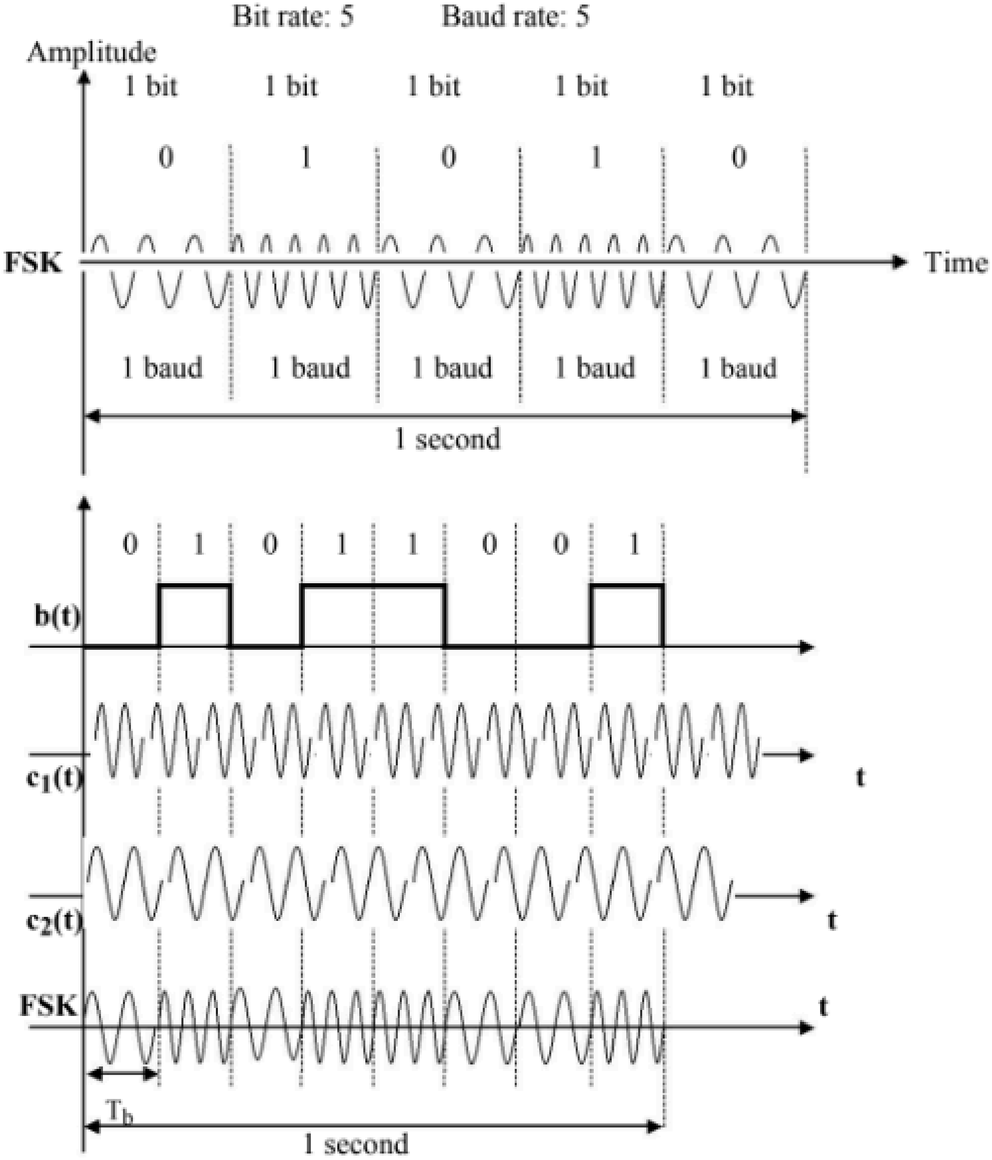
In ASK the strength of the signal is varied to represent binary 1 or 0. Both frequency and phase remain constant, while the amplitude changes.



Bit duration is the period of time that defines one bit. The peak amplitude of the signal, during each bit duration, is constant and its value depends on the bit (0 or 1). The transmission speed using ASK is limited by the physical characteristics of the transmission medium.

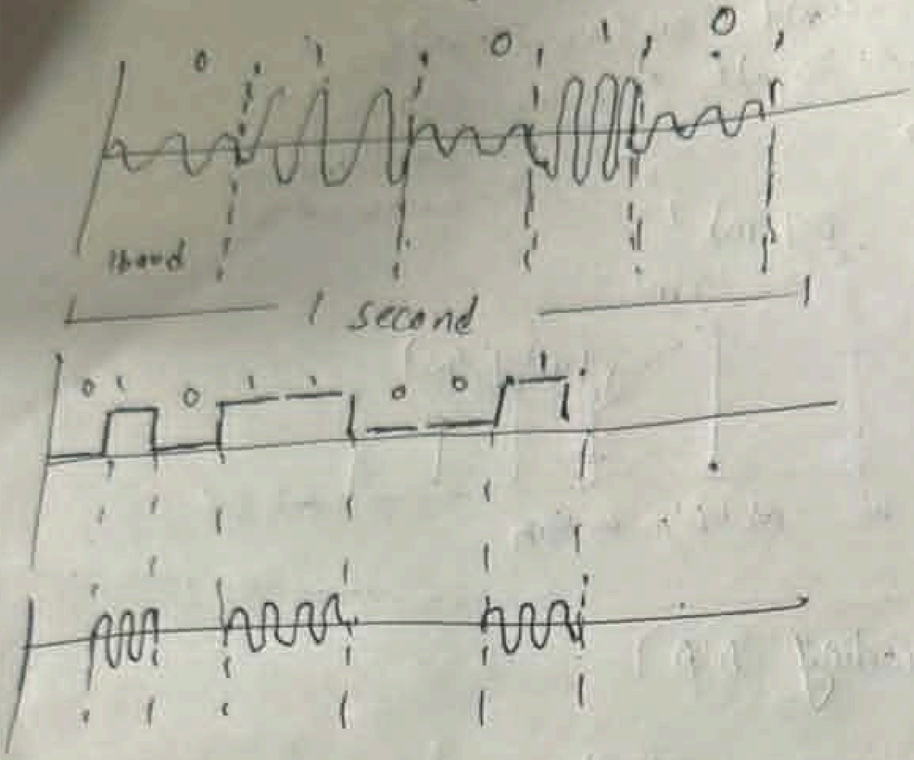
Frequency Shift Keying (FSK)

In frequency shift keying (FSK), the frequency of the signal is varied to represent binary 1 or 0. The frequency of the signal during each bit duration is constant and its value depends on the bit (0 or 1): both peak amplitude and phase remain constant.

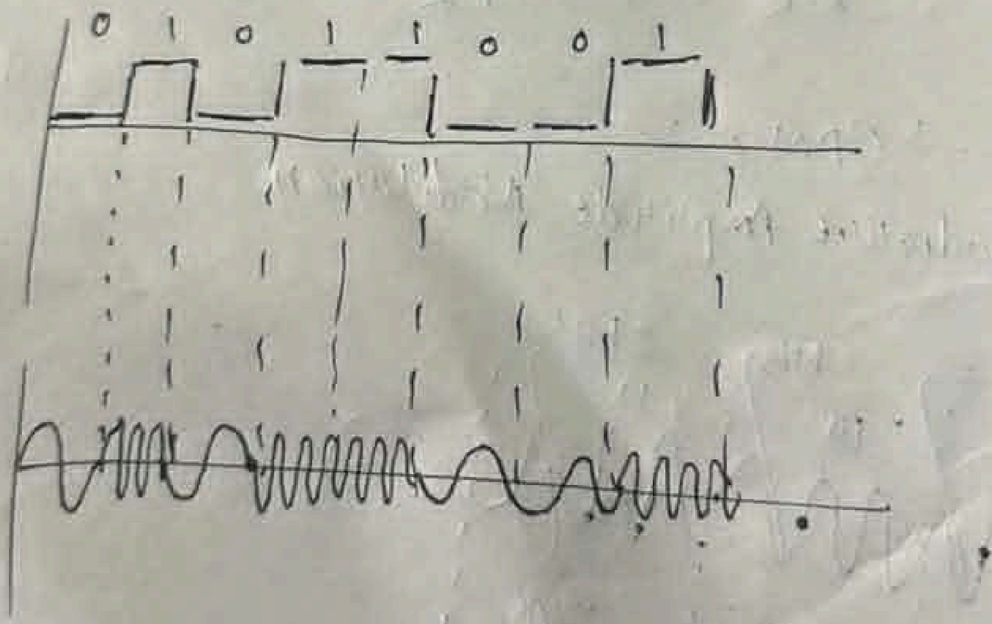


Source DC Notes

* Amplitude Shift Keying.

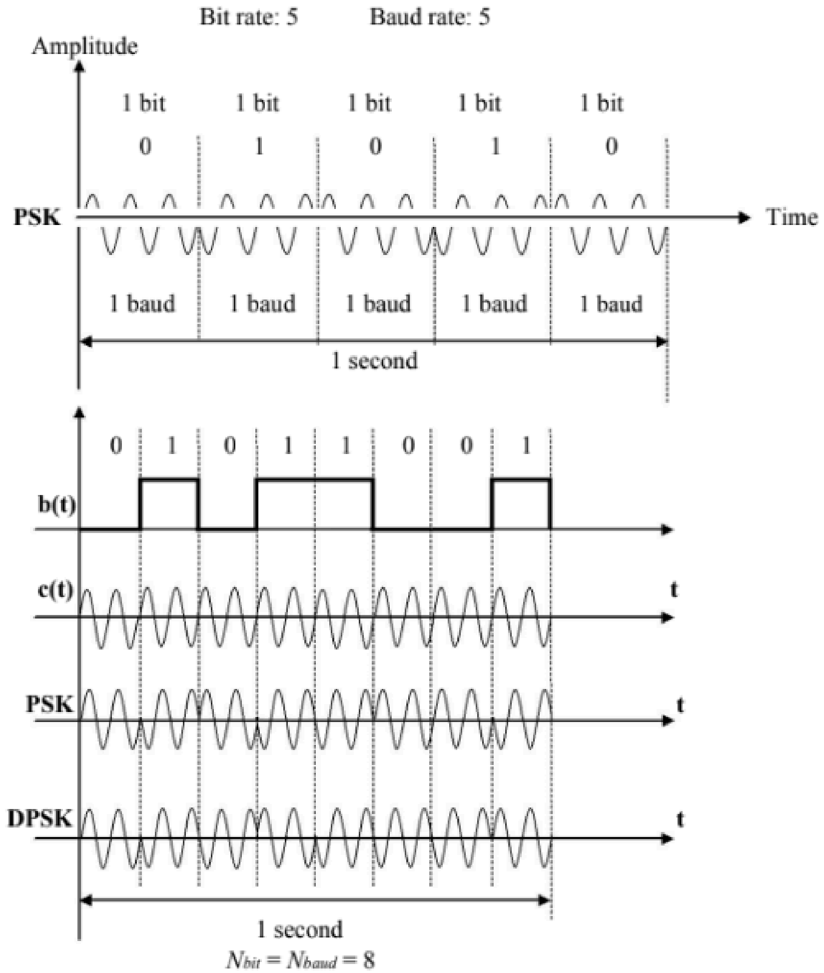


* Frequency Shift Keying



Phase Shift Keying (PSK)

In the PSK, the phase is varied to represent binary 1 or 0. Both peak amplitude and frequency remain constant as the phase changes. The phase of the signal during each bit duration, is constant and its value depends on the bit (0 or 1).

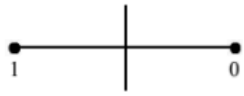


DPSK eliminates the need for a coherent reference signal at the receiver by combing two basic operations at the transmitter:

[Most IMP]

PSK Constellation

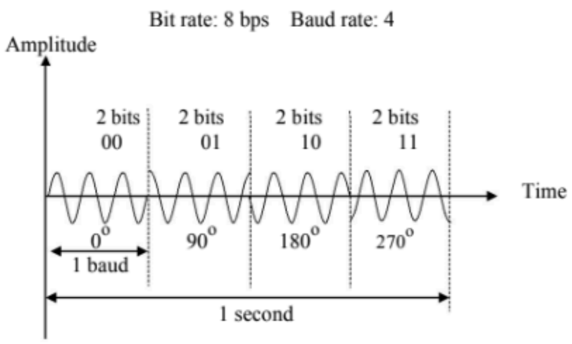
Bit	Phase
0	0^{0}
1	180°



Constellation diagram

The above method is often called 2-PSK, or binary PSK, because two different phases (0° and 180°) are used in the encoding.

PSK is not susceptible (easily influenced) to the noise degradation that affects ASK, nor to the bandwidth limitations of FSK. This means that smaller variations in the signal can be detected reliably by the receiver. Therefore instead of utilising only two variations of a signal, each representing one bit, we can use four variations and let each phase shift represent two bits.



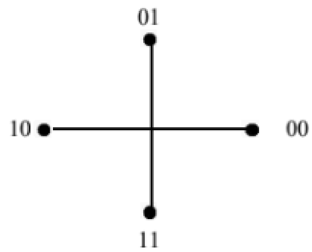
4-PSK (Quadrature-PSK)

This technique is called 4-PSK or Q-PSK. The pair of bits represented by each phase is called a dibit.

Data can be transmitted two times as fast using 4-PSK as using 2-PSK.

4–PSK characteristics

Dibit	Phase
00	0^{0}
01	90 °
10	180^{0}
11	270^{0}

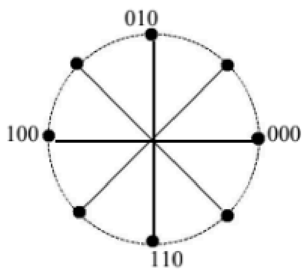


Constellation diagram

8–PSK Characteristics

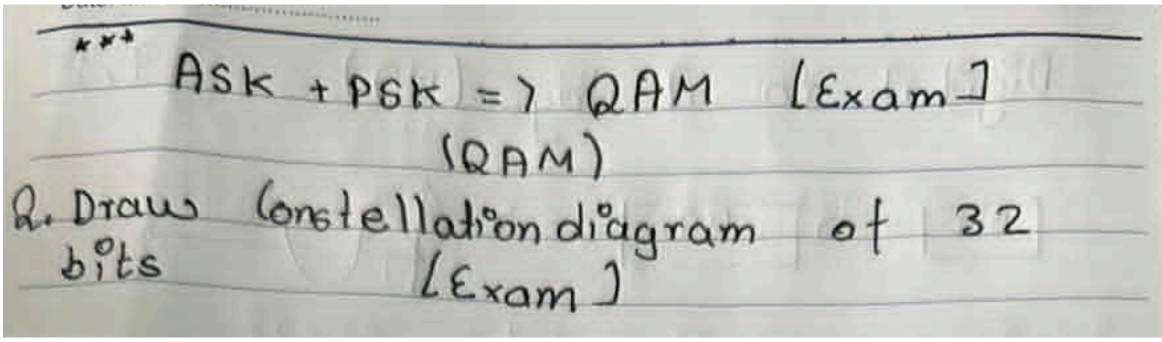
Tribit	Phase
000	0°
001	45^{0}
010	90^{0}
011	135^{0}
100	180^{0}
101	225^{0}
110	270^{0}
111	315^{0}

Bit rate of 8-PSk is three as that of 2-PSK



Constellation diagram

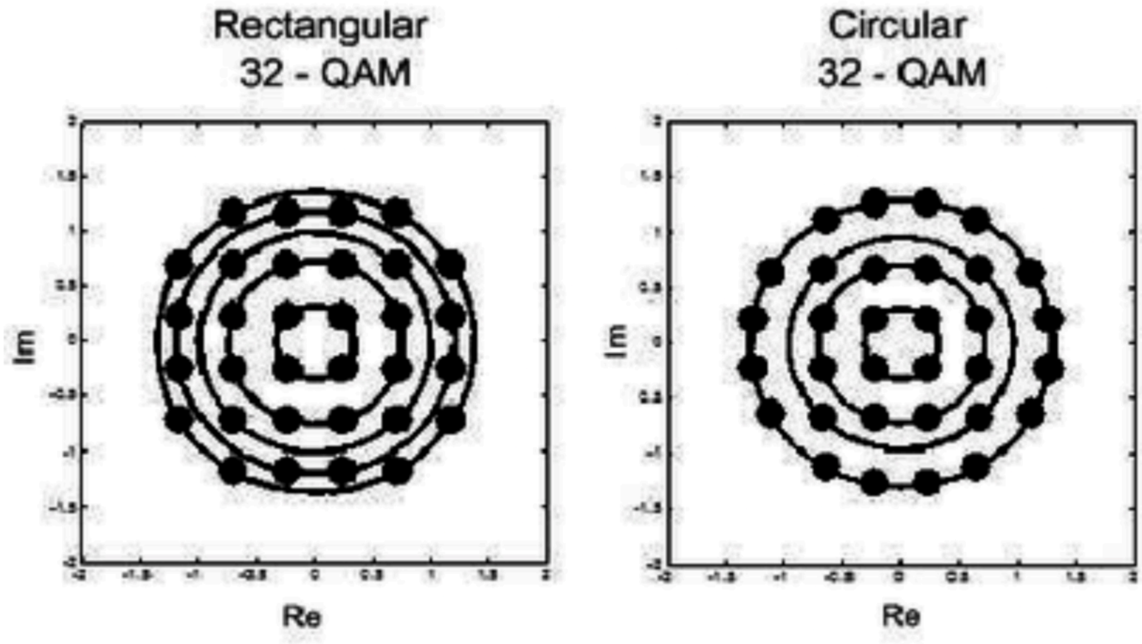
[Most Probable Questions]



*** ASK + PSK => QAM [Exam]
(QAM)

Q. Draw Constellation diagram of 32 bits [Exam]

Q. Prepare a constellation diagram of 32-QAM. [2082 Kartik BEI] [2081 Chaitra BEI] [2081 Ashadh BEI] [2080 Chaitra BEI]



https://www.researchgate.net/profile/Huseyin-Arslan-4/publication/228796649/figure/fig1/AS:300904025739264@1448752518193/Different-Constellation-Mappings-for-QAM-orders-of-16-32-and-64.png